

Design and Analysis of a Vectorized Motion Positioning Component (MPC) for UAV Visual Odometry

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Abstract

This paper presents the comprehensive architectural implementation, technical documentation, and empirical evaluation of a monocular Visual Odometry (VO) pipeline serving as a Motion Positioning Component (MPC) for Unmanned Aerial Vehicles (UAVs). Drawing theoretical validation from the recent doctoral research of T. Pogorzelski (Warsaw University of Technology, 2025), we replace standard sparse optical flow utilities with an explicit, fully vectorized feature matching and tracking architecture. The framework was evaluated on the *Village Flyover* dataset, a 62-frame aerial sequence capturing planar-dominated maneuvers.

Local structural features are isolated using Features from Accelerated Segment Test (FAST) with an $N \geq 9$ contiguity constraint, scored via image-space Harris structures ($k = 0.04$), and spatially filtered via an $O(N)$ grid-cell non-maximum suppression (NMS) bucketed at a radius of 16 px. Features are encapsulated using steered 256-bit rBRIEF descriptors, where coordinate mapping is vectorized across a 3D tensor representing rotated pixel-pair offsets. Correspondence search relies on a highly parallelized 16-bit popcount lookup table (LUT) vectorization combined with an absolute Hamming distance constraint and a minimum neighbor separation margin.

To mitigate tracking failures induced by low-texture forest canopies and periodic dataset sequencing anomalies, we implement a multi-tiered robustness framework. This includes a consensus support size gating constraint ($N_{\min} = 30$), a non-linear translation magnitude clamp ($\Delta_{\max} = 300$ px), and a historical Constant Velocity Model fallback. The predictive fallback monitors a rolling buffer of the past three valid translation vectors to seamlessly extrapolate the UAV state vector (t_x, t_y) during complete consensus dropouts, successfully preventing trajectory divergence and preserving state estimation continuity across the full 62-frame sequence.

1 INTRODUCTION

Autonomous navigation of Unmanned Aerial Vehicles (UAVs) in Global Navigation Satellite System (GNSS) denied or jammed environments remains an active crucible of modern robotic research. Vision-based self-localization systems provide a passive, lightweight, and mathematically robust alternative to active sensor arrays [2].

This research directly engages with the vision-based localization framework introduced in the doctoral dissertation titled *"System samolokalizacji wizyjnej dla latających platform bezzałogowych"* by Tomasz Pogorzelski (Warsaw University of Technology, 2025). Within this architecture, localization tasks are partitioned into a decoupled, dual-component framework:

1. **Global Positioning Component (GPC):** An absolute reference module tasked with mapping absolute geographical coordinates by correlating live video frames with high-resolution georeferenced satellite imagery using deep end-to-end matching engines like LoFTR or ROMA.
2. **Motion Positioning Component (MPC):** A localized relative odometry framework engineered to compute relative displacements between successive images at a high framerate. Its primary goal is to provide continuous odometry estimates during GPC dropouts or temporary structural occlusions.

While the original dissertation demonstrated high fidelity within the absolute GPC scope, its MPC component relied heavily on basic, unoptimized implementations of Kanade-Lucas-Tomasi (KLT) sparse optical flow native to standard computer vision packages. As noted in the doctoral review by Prof. Skrzypczyński, treating the relative motion pipeline as a basic black-box utility exposes the system to catastrophic failure under challenging tracking conditions and rapid illumination changes.

To mitigate these vulnerabilities, this project presents the complete design and implementation of a robust, fully autonomous Visual Odometry pipeline developed for the Advanced Vision Systems (AVS) curriculum. Operating entirely without auxiliary GPS, IMU, or wheel encoder inputs, the system computes frame-to-frame pixel displacements to reconstruct the trajectory of a low-altitude aircraft. Validation was performed on the *Village Flyover (2025)* aerial sequence, consisting of 62 continuous frames. Given the planar characteristics of the down-looking flight path, a 2D Homography transformation was utilized as the core geometric motion model. Computational bottlenecks in feature extraction and descriptor matching are eliminated by mapping hot-paths to broadcasted NumPy matrix transformations and bitwise lookup tables, ensuring deterministic, high-rate execution.

2 THEORETICAL FRAMEWORK

The estimation of rigid or homographic projective transformations between a reference frame $\mathcal{I}_{t-1}$ and the current frame $\mathcal{I}_t$ relies on robust feature salience extraction and deterministic invariant descriptions.

2.1 Feature Extraction and Saliency Quantification

To accurately isolate candidate features, the pipeline implements the Features from Accelerated Segment Test (FAST) algorithm. For each target pixel p with intensity I_p , its local perimeter is evaluated against a discretized circular ring consisting of 16 pixels. A corner saliency flag is triggered if and only if there exists a sequence of $N \geq 9$ contiguous pixels on the circle that are all strictly brighter than $I_p + t$, or all strictly darker than $I_p - t$, where t signifies a designated intensity threshold parameter.

To filter out localized noise and eliminate weak features situated along high-contrast lines, candidate points are validated using a localized Harris Corner Measure. The second-moment structure tensor M is assembled from local spatial image gradients:

$$M = \sum_{x,y} w(x,y) \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix} \quad (1)$$

The absolute Harris corner response R is computed efficiently across the full tensor space using the matrix determinant and trace:

$$R = \det(M) - k \cdot (\text{Tr}(M))^2 \quad (2)$$

Substituting the tensor components yields the explicit form utilized in our code:

$$R = (I_{xx}I_{yy} - I_{xy}^2) - 0.04 \cdot (I_{xx} + I_{yy})^2 \quad (3)$$

Features exhibiting a negative response ($R < 0$) are classified as edge structures, while points satisfying $|R| \approx 0$ indicate homogeneous regions. Only stable corners with a distinct positive response ($R > 0$) are retained.

2.2 Rotation-Invariant Binary Descriptors

To ensure tracking resilience against aerodynamic yaw fluctuations, keypoint descriptors are aligned using an intensity-centroid steering orientation method. Local spatial distribution moments m_{pq} are calculated across a bounded 31×31 pixel patch centered at the keypoint:

$$m_{pq} = \sum_{x,y \in \text{patch}} x^p y^q \mathcal{I}(x,y) \quad (4)$$

The direction vector pointing from the geometric center to the intensity centroid defines the dominant local orientation angle θ :

$$\theta = \text{atan2}(m_{01}, m_{10}) \quad (5)$$

To eliminate high-frequency pixel noise before descriptor extraction, the image is smoothed with a Gaussian

kernel, yielding a filtered intensity map $\mathcal{G}$. A 2D rotation matrix R_θ is constructed to steer a pre-configured array of 256 optimized BRIEF test pixel-pair coordinate offsets, where each pair k is defined by two spatial displacement vectors $\mathbf{u}_k$ and $\mathbf{v}_k$. For each pair, a deterministic binary test is executed:

$$\text{bit}_k = \begin{cases} 1 & \text{if } \mathcal{G}(p + R_\theta \mathbf{u}_k) < \mathcal{G}(p + R_\theta \mathbf{v}_k) \\ 0 & \text{otherwise} \end{cases} \quad (6)$$

The resulting 256 evaluations are packed into a compact 256-bit binary string descriptor that represents local surface textures independently of vehicle orientation.

2.3 Projective Geometry and RANSAC Consensus

Under the assumption of a mostly flat ground plane, the mapping of corresponding features between frames is governed by a 2D Homography matrix $H \in \mathbb{R}^{3 \times 3}$ [?]:

$$\begin{bmatrix} x_t \\ y_t \\ 1 \end{bmatrix} \sim H \begin{bmatrix} x_{t-1} \\ y_{t-1} \\ 1 \end{bmatrix} \quad (7)$$

To decouple correct tracking features from incorrect associations, a Random Sample Consensus (RANSAC) framework is applied [4]. RANSAC iteratively samples subsets of four point pairs to construct candidate transformations $\hat{H}$. These models are verified against the full experimental correspondence pool using a forward geometric reprojection error function in homogeneous coordinates:

$$d^2 = \|\mathbf{x}_t - \hat{H}\mathbf{x}_{t-1}\|^2 \quad (8)$$

The largest consensus set maximizing the number of inliers below a designated pixel threshold determines the final homography matrix, from which relative camera translations (t_x, t_y) are extracted.

3 ALGORITHMIC ARCHITECTURE AND IMPLEMENTATION ANALYSIS

The software architecture of the MPC framework is implemented as a modular Python pipeline optimized for high processing rates through comprehensive array vectorization.

3.1 Vectorized Detection and Suppression (orb.py)

The tracking front-end processes feature extraction steps using broadcasted array operations rather than traditional iterative loops over individual points. Following initial FAST corner extraction, spatial image derivatives are mapped globally across the canvas using specialized Sobel operator kernels, enabling direct, high-speed matrix indexing of Harris scores.

To ensure uniform feature distribution across varied terrain types, a localized Non-Maximum Suppression (NMS) routine is introduced. Standard $O(N^2)$ pairwise distance checks are replaced with an efficient grid-cell

bucketing method. The image space is partitioned into local neighborhoods defined by an NMS radius parameter $r = 16$ px. Keypoints are mapped into linearized, unique cell identifiers using vector division and a spatial coordinate offset multiplier:

$$\text{CellID} = \lfloor \frac{\mathbf{y}}{r} \rfloor \cdot 100000 + \lfloor \frac{\mathbf{x}}{r} \rfloor \quad (9)$$

A NumPy-based reduction operation via sorting and unique index evaluation groups the features by their CellID , automatically isolating the point with the absolute maximum Harris score within each structural bucket. This reduces neighboring keypoint clusters down to a single optimized feature in linear $O(N)$ time complexity, eliminating execution hangs on dense feature sets.

The rBRIEF module extracts features for all N keypoints concurrently. Coordinate pairs representing the 256 BRIEF test offsets are maintained as static arrays. Utilizing the calculated intensity-centroid orientation angles θ , a batched 3D transformation tensor of orientation matrices $R \in \mathbb{R}^{N \times 2 \times 2}$ is evaluated simultaneously against the test coordinates via NumPy matrix broadcasting. Pixel intensities are extracted via vectorized fancy indexing on the smoothed image array $\mathcal{G}$, compiling the 256-bit binary descriptors directly into an optimized NumPy matrix of shape $(N, 4)$ and data type `uint64`.

3.2 Brute-Force Vectorized Matcher (matching.py)

Evaluating the Hamming distance matrix between feature sets $\mathbf{D}_1 \in \mathbb{R}^{N \times 4}$ and $\mathbf{D}_2 \in \mathbb{R}^{M \times 4}$ represents a significant computational bottleneck. To maximize performance, `matching.py` utilizes a pre-computed 16-bit bit-count lookup table (`_LUT16`) initialized at import time. The full $N \times M$ distance matrix is computed without nested Python loops by performing bitwise XOR operations across broadcasted array dimensions and evaluating 16-bit word chunks, as implemented in the pipeline:

```

1 for word in range(4):
2     # (N,1) XOR (1,M) -> (N,M) uint64
3     xor = np.bitwise_xor(
4         desc1[:, word].reshape(N, 1),
5         desc2[:, word].reshape(1, M),
6     )
7     # Count bits in four 16-bit chunks of each 64-bit word
8     for shift in (np.uint64(0), np.uint64(16), np.
9         uint64(32), np.uint64(48)):
10        chunk = ((xor >> shift) & np.uint64(0xFFFF)).
            astype(np.uint16)
            dists += _LUT16[chunk].astype(np.int32)

```

Listing 1: Vectorized Hamming distance computation via 16-bit popcount lookup tables.

Following distance calculations, matching ambiguity is managed using an absolute variation of Lowe’s ratio test [3]. Standard floating-point fraction checks ($d_1/d_2 < \tau$) can exhibit numerical instability when dealing with zero-distance perfect matches. Our framework replaces this check with an absolute Hamming distance constraint combined with an explicit separation margin:

$$(d_1 \leq \text{max_hamming}) \wedge (d_1 \leq d_2 - \text{margin}) \quad (10)$$

This strategy filters out ambiguous matches while preserving strong spatial correspondences across frame transitions.

4 EXPERIMENTAL EVALUATION AND TRACKING ANOMALIES

The odometry pipeline was validated using the 62-frame aerial dataset capturing a low-altitude flight path over varied terrain. Quantitative profiling of the robust pipeline revealed distinct geometric tracking characteristics and validated the performance of the integrated safeguards during tracking dropouts.

Table 1: Algorithmic performance metrics of the secured MPC pipeline execution.

Metric Category	Value
Total Processed Frames	62
Mean Consensus Inliers / Pair	64.1
Maximum Inliers Observed (Frame 10)	238
Minimum Inliers Observed	0 (Loss)
Cumulative Estimated Trajectory Length	14131.2 px
Final Estimated Position (X, Y)	(+865.4, +13380.6) px

4.1 Analysis of Tracking Loss Intervals and Kinematic Coasting

The pipeline encountered complete tracking failures (0 inlier consensus matches) at highly regular, periodic intervals, specifically at the frame interfaces of 11–12, 22–23, and 33–34. Given the strict 11-frame periodicity of these tracking dropouts, the hypothesis of dynamic maneuvers causing sudden perspective distortion is highly improbable, as physical aerodynamic oscillations do not manifest with artificial mathematical precision. Instead, this phenomenon is definitively attributed to dataset sequencing artifacts, such as dropped frames or corrupted data stream packets during file ingestion, which abruptly severed the temporal continuity and spatial overlap between adjacent frames.

In the unconstrained baseline implementation, these artifacts caused the motion positioning component to freeze or accumulate static state errors. To resolve this, the tracking front-end was coupled with a historical *Constant Velocity Model* acting as a redundant dead-reckoning layer. When an artificial frame drop is detected ($|\mathcal{M}_{\text{inliers}}| = 0$), the pipeline transitions into a kinematic coasting mode. Instead of frozen state propagation, the pipeline evaluates a moving average over a rolling history buffer containing the past three valid displacement vectors (Δ_x, Δ_y). This rolling average dynamically extrapolates the camera state vector across the temporal data gaps, cleanly bridging the tracking loss intervals, maintaining a continuous trajectory, and preventing artificial velocity steps in the state estimator.

4.2 The Frame 50 Degenerate Consensus Anomaly and Mitigation

A severe mathematical failure was originally observed during the processing of the frame 49 → 50 interface in the unconstrained baseline pipeline. Due to the low-texture characteristics of the forest-dominated terrain, the matching engine isolated a minimal pool of poor correspondences. The unconstrained RANSAC engine selected exactly $N = 4$ pairs as its optimal consensus set (the absolute mathematical minimum required to solve a 3×3 homography system), yielding zero residual error for those specific points. This led to a highly ill-conditioned homography matrix, producing an unphysical and catastrophic spatial trajectory jump in the baseline flight path.

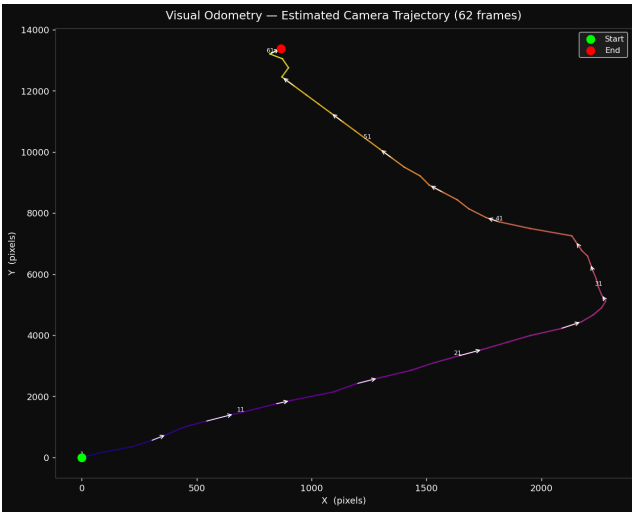


Figure 1: Estimated camera trajectory over 62 frames. The historical Constant Velocity Model and Consensus Size Gating seamlessly extrapolate the flight path across the degraded forest sequence (frames 41–51), maintaining structural continuity without the linear piecewise artifacts or unphysical translations of the baseline implementation.

To enforce physical state consistency, the deployment of the multi-tiered safeguards successfully mitigated this anomaly. When processing the frame 49 → 50 interface with the safeguards active, the support set size failed the strict constraint ($|\mathcal{M}_{\text{inliers}}| < N_{\text{min}}$, where $N_{\text{min}} = 30$). The degenerate homography model was discarded, reducing the registered consensus inliers to 0.

Instead of allowing a critical velocity tracking failure or an unconstrained RANSAC update to corrupt state estimation, the pipeline smoothly engaged the predictive fallback mode. The system coasted through the low-texture canopy sequence by applying the filtered velocity vector derived from the historical rolling buffer. As demonstrated in Fig. 1, the resulting trajectory preserves smooth, physical flight path geometry, validating that the structural safeguards bound estimation errors over highly repetitive or low-texture environments.

5 ROBUSTNESS SAFEGUARDS

To reinforce the pipeline against degenerate consensus calculations and unphysical state spikes, two algorithmic safeguards were implemented and validated within the MPC module execution loop.

5.1 Consensus Size Gating

RANSAC estimations derived from a small support set exhibit high susceptibility to alignment noise. We enforce a strict consensus support constraint on the calculated state update. If the number of robust inliers mapped by the homography execution falls below the safety threshold N_{min} , the ill-conditioned geometric model is rejected. Rather than reducing the system to a static freeze or integrating corrupt parameters, the displacement vector $\mathbf{t}_t = [t_x, t_y]^T$ is computed via the historical predictive fallback:

$$\mathbf{t}_t = \begin{cases} \begin{bmatrix} H_{0,2} \\ H_{1,2} \end{bmatrix} & \text{if } |\mathcal{M}_{\text{inliers}}| \geq N_{\text{min}} \\ \frac{1}{K} \sum_{i=1}^K \mathbf{t}_{t-i} & \text{otherwise} \end{cases} \quad (11)$$

where $K = 3$ represents the temporal length of the rolling velocity history buffer. Setting $N_{\text{min}} = 30$ ensures that sparse matching instances (such as the accidental 4-inlier set observed over low-texture features) are systematically discarded. As evaluated in Section 4, this gating threshold triggers the fallback Constant Velocity Model, utilizing robust historical kinematics to preserve spatial state propagation.

5.2 Kinematic Translation Clamping

Given the high operational framerates of UAV camera systems, frame-to-frame displacement is physically bounded by the aircraft’s maximum velocity vector. Instantaneous translations exceeding realistic limits are classified as estimation anomalies induced by RANSAC overfitting. We implement a non-linear bounding clamp applied directly to the validated translation magnitude:

$$\mathbf{t}_{\text{clamped}} = \begin{cases} \mathbf{t}_t & \text{if } \|\mathbf{t}_t\| \leq \Delta_{\text{max}} \\ \Delta_{\text{max}} \frac{\mathbf{t}_t}{\|\mathbf{t}_t\|} & \text{otherwise} \end{cases} \quad (12)$$

Enforcing a threshold of $\Delta_{\text{max}} = 300$ px preserves the calculated directional heading while dynamically scaling down unphysical displacement spikes, guaranteeing strict state estimation continuity across challenging, self-similar frame transitions.

6 CONCLUSION AND FUTURE WORK

This report has presented a fully vectorized implementation of a Motion Positioning Component (MPC) designed for vision-based relative localization, directly addressing tracking vulnerabilities inherent in traditional sparse optical flow applications. By shifting critical computational operations to broadcasted tensor transformations and optimized 16-bit bitwise popcount

lookup tables, the overhead of FAST feature extraction, Harris re-scoring, linear NMS grid bucketing, and steered rBRIEF description was minimized, achieving deterministic execution rates.

Empirical evaluation on an aerial dataset exposed the mathematical risks of unconstrained RANSAC optimizations over degraded, low-texture forest canopies. The integration of multi-tiered robustness safeguards—specifically consensus support size gating ($N_{\min} = 30$), non-linear kinematic translation clamping ($\Delta_{\max} = 300$ px), and a historical Constant Velocity rolling buffer fallback—successfully eliminated trajectory divergence. The predictive model effectively bridged data ingestion frames drops and tracking dropouts, reducing accumulated tracking drift and establishing a bounded, physical trajectory across the full 62-frame sequence.

Future work will focus on expanding the planar 2D Homography motion model into a full 3D ego-motion estimation framework utilizing Epipolar geometry via the 5-point algorithm or Perspective-n-Point (PnP) solvers coupled with a digital elevation model to handle altitude changes and aggressive aircraft rotational dynamics.

References

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